EduNavigator- CCNY Intelligent Academic Navigation System Software Requirements Specification For <Subsystem or Feature>

Version <1.0>

Revision History

Dat	Version	Description	Author
03/23/2026	<1.0>	EduNavigator – CCNY Intelligent	Mim D., Zara R.,

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Software Requirements Specification

1. Introduction

1.1 Purpose - Guillermo

The functional and non-functional needs for EduNavigator, an AI-enabled college program management system, are outlined in this software requirements specification. Registrars, instructors, students, and visitors are the four categories of users for which the system is designed to support academic and administrative functions in a small college setting. Semester management, admissions, course registration, grading, complaints, reviews, warnings, suspensions, graduation processing, and an AI assistant with access to local college data are all included in the system.

1.2 Scope – Guillermo

A desktop program or local graphical user interface (GUI) may be used to construct EduNavigator, a toy but completely consistent college administration system. For demonstration reasons, the system must accommodate about ten students and stay within a single primary GUI window. It implements college standards automatically wherever feasible and automates academic process over the course of four semester periods. Additionally, it has an AI-powered question section supported by local data and backup LLM answers with warnings against hallucinations.

1.3 Definitions, Acronyms, and Abbreviations – Guillermo

- **Registrar**: Super user with full system access
- **Instructor**: Faculty member assigned to classes by registrars
- **Student**: Approved enrolled user who manages course activity
- **Visitor**: Public user who can browse information and apply
- **GPA**: Grade Point Average
- **LLM**: Large Language Model
- **Vector DB**: Database used to store local knowledge embeddings
- **Honor Roll**: Academic distinction automatically assigned by GPA rule
- **Wait-list**: Queue for full classes
- **Taboo Words**: Restricted words configured by registrars for review moderation

1.4 References – Guillermo

- User project specification document for College0 requirements.
- Software Requirements Specification (SRS) Template
- **IEEE Std 830-1998** – *IEEE Recommended Practice for Software Requirements Specifications*
- Ian Sommerville, *Software Engineering (10th Edition)*

1.5 Overview – Mim

This Software Requirements Specification describes the requirements for the EduNavigator – CCNY Intelligent Academic Navigation System. The document is organized into four main sections. Section 1 introduces the purpose, scope, definitions, and references of the system. Section 2 provides an overall description of the system, including its structure, user roles, and operating environment. Section 3 presents the detailed functional and non-functional requirements through use-case reports and supplementary requirements. Section 4 includes supporting information such as system interface descriptions, storyboards, and additional documentation that assist in understanding and implementing the system.

2. Overall Description – Mim

EduNavigator is an AI-enabled academic management system designed to support administrative and academic processes within a small-scale college environment such as CCNY. The system serves as a centralized platform that integrates student management, course registration, grading, reviews, complaints, and graduation tracking into a single interface.

From a product perspective, EduNavigator functions as a standalone desktop or local GUI-based application that operates within a single window for consistency and ease of use. It is designed as a simplified system for demonstration purposes, supporting a limited number of users and courses while maintaining realistic academic rules and workflows.

The system provides several key functions, including student and instructor application processing, course registration with conflict detection, waitlist management, grading, GPA calculation, review and rating systems, complaint handling, warning and suspension tracking, and graduation eligibility verification. These features are organized across four academic phases: class setup, course registration, class running, and grading.

EduNavigator supports four main user roles: visitors, students, instructors, and registrars. Visitors can browse general information and apply to join the system. Students can manage courses, reviews, and academic progress. Instructors are responsible for managing classes, grading students, and handling waitlists. Registrars act as administrators with full control over the system, including approvals, system rules, and disciplinary actions.

The system operates under several constraints. It must maintain a single-window graphical user interface, support a small dataset (approximately ten students), and enforce academic rules such as course limits, GPA thresholds, and warning policies. Additionally, the system must ensure role-based access control and data consistency.

EduNavigator also depends on an AI-powered assistant feature that enhances user interaction. The AI system first retrieves answers from a local vector database containing program-specific knowledge. If no relevant information is found, the system forwards the query to an external large language model and displays a warning about possible inaccuracies.

Overall, EduNavigator is designed to simulate a real academic management system while incorporating intelligent features to improve usability, automation, and decision-making.

2.1 Use-Case Model Survey - Mashiyat

The **EduNavigator system** supports four main types of users: **Visitor, Student, Instructor, and Registrar**. Each actor interacts with the system to perform different academic and administrative tasks.

Visitor:
A visitor can access general information about the EduNavigator program, including the program introduction, highest-rated classes, lowest-rated classes, and students with the highest GPA. Visitors can also apply to become students or instructors and use the AI question-answering feature.

Student:
A student is an approved user who can log into the system and manage academic activities. Students

can register for courses during the registration period, view their academic records, write reviews for classes, submit complaints, apply for graduation, and ask questions using the AI system.

Instructor:
An instructor is assigned to courses by the registrar. Instructors can view their assigned classes,

access student records, manage waitlists, assign grades during the grading period, submit complaints, and use the AI feature for additional information.

Registrar:
The registrar is the system administrator with full access. The registrar manages student and instructor applications, sets up courses and semester periods, assigns instructors, processes complaints, manages warnings and suspensions, controls taboo words, and approves graduation applications.

Major Use Cases:

The EduNavigator system includes the following main use cases:

- View Program Information
- Apply to Become a Student
- Apply to Become an Instructor
- Register for Courses
- Manage Waitlist
- Assign Grades
- Write Class Review
- Submit Complaint
- Apply for Graduation
- Ask AI Questions

These use cases represent the primary interactions between users and the system.

2.2 Assumptions and Dependencies – Guillermo

The system shall:

- display college introduction, highest-rated classes, lowest-rated classes, and highest-GPA students to all users
- allow visitors to apply as students or instructors
- let registrars approve or reject applications
- assign student IDs and first-login password change requirements
- manage semester phases:
    - class setup period

    - course registration period
    - class running period
    - grading period
- enforce course registration limits of 2 to 4 courses
- detect time conflicts
- manage class capacity and wait-lists
- allow retakes only when previous result was F
- cancel under-enrolled classes
- trigger special registration after cancellation
- manage instructor warnings and suspension rules
- allow student reviews and star ratings
- censor taboo words in reviews
- apply student and instructor warnings
- allow grading and GPA calculations
- terminate or warn students based on GPA and repeated failures
- process graduation applications
- handle complaints
- provide role-based dashboards
- provide an AI question-answer feature with local data and LLM fallback

2.3 User Classes and Characteristics - Guillermo

Visitors

- can browse public information
- can apply to become students or instructors

Students

- can view their own records
- register for courses
- review classes before grades are posted

- apply for graduation
- submit complaints

Instructors

- can access their assigned classes
- can see academic records of students in their classes
- can admit wait-listed students
- assign grades
- submit complaints against students

Registrars

- have full access
- manage applications, semester setup, class offerings, warnings, complaints, and graduation decisions

3. Specific Requirements – Zara

This section defines the detailed functional and non-functional requirements of the EduNavigator system. These requirements are specified at a level that allows developers to design and implement the system, and enables testers to verify that the system meets its intended behavior. The requirements are primarily captured through use-case reports, which describe interactions between users (Visitor, Student, Instructor, and Registrar) and the system. Additional system-wide behaviors, constraints, and performance expectations are described in the supplementary requirements. Together, these specifications ensure that the system supports academic management, user interactions, and AI-assisted features as outlined in the overall system description.

3.1 Use-Case Reports- Mashiyat

Use Case 1 – Apply to Become a Student

Actor: Visitor

Description:

A visitor applies to become a student in the EduNavigator system.

Preconditions:
The user is a visitor.

Main Flow:

1. The visitor selects “Apply as Student”.
2. The visitor enters required information including GPA.
3. The system records the application.
4. The registrar reviews the application.
5. If GPA > 3.0 and quota is not reached, the student is accepted.
6. The system generates a student ID and temporary password.

Alternative Flow:
If the registrar rejects a valid applicant, justification must be provided.

Postconditions:
The applicant is accepted or rejected.

Use Case 2 – Apply to Become an Instructor

Actor: Visitor

Description:
A visitor applies to become an instructor in the EduNavigator system.

Preconditions:
The user is a visitor.

Main Flow:

1. Visitor selects “Apply as Instructor”.
2. Visitor enters qualifications.
3. The system stores the application.
4. Registrar reviews and approves or rejects.

Postconditions:
Approved instructors are assigned courses.

Use Case 3 – Register for Courses

Actor: Student

Description:
A student registers for courses during the registration period in EduNavigator.

Preconditions:
The system is in the course registration period.

Main Flow:

1. Student logs into the system.
2. Student views available courses.
3. Student selects 2–4 courses.
4. System checks for time conflicts.
5. System checks class capacity.
6. If space is available, student is enrolled.
7. If full, student is added to the waitlist.

Postconditions:
Student is enrolled or waitlisted.

Use Case 4 – Assign Grades

Actor: Instructor

Description:
An instructor assigns grades to students in EduNavigator.

Preconditions:
The system is in the grading period.

Main Flow:

1. Instructor logs into the system.
2. Instructor selects a course.
3. Instructor assigns grades to students.
4. The system records grades.

Postconditions:
Grades are stored in the system.

Use Case 5 – Write Class Review

Actor: Student

Description:
A student writes a review for a course in EduNavigator.

Preconditions:
Student is enrolled and grades are not yet posted.

Main Flow:

1. Student selects a course.
2. Student writes a review and assigns a rating.
3. System checks for taboo words.
4. If 1–2 taboo words → replace with “*” and give warning.
5. If ≥3 taboo words → reject review and give 2 warnings.
6. System stores the review.

Postconditions:
Review is saved or rejected.

Use Case 6 – Submit Complaint

Actor: Student / Instructor

Description:
A user submits a complaint in EduNavigator.

Preconditions:
User is logged in.

Main Flow:

1. User submits complaint.
2. Registrar reviews complaint.
3. Registrar takes action (warning or rejection).

Postconditions:
Complaint is resolved.

Use Case 7 – Apply for Graduation

Actor: Student

Description:
A student applies for graduation in EduNavigator.

Preconditions:
Student has completed 8 courses.

Main Flow:

1. Student submits graduation application.
2. System checks academic records.
3. Registrar reviews application.
4. If requirements are met, student graduates.

Alternative Flow:
If requirements are not met, the student receives a warning.

Postconditions:
Student graduates or receives a warning.

Use Case 8 – Ask AI Question

Actor: Visitor, Student, Instructor

Description:
A user asks a question using the AI feature in EduNavigator.

Preconditions:
User is using the AI interface.

Main Flow:

1. User enters a question.
2. System searches local database.
3. If found → display answer.
4. If not → send to LLM.
5. Display answer with warning.

Postconditions:
User receives a response.

3.2 Supplementary Requirements- David

User Roles

The roles the system will use:

- Registrars: Users that have full access to all system functions.
- Instructors: User that can view the course that they have been assigned by the registrars.
- Students: Users that can view their academic records and manage their course registrations.
- Visitors: Users that can see public information about the students or the courses that are available and apply to be a student or an instructor.

(GUI)- Graphical User Interface

The interface that is available for all users will include:

- A general introduction to the program.
- The highest rated classes.
- The lowest rated classes.
- Students with the highest GPA.

Academic Period Management

The semesters will be divided into four periods that is managed by the registrar:

1. **Class Setup Period**

2. **Course Registration Period**
3. **Class Running Period**
4. **Grading Period**

Course Registration Restrictions

Students need to register for 2-4 courses during the registration period. Other restrictions the system will have:

- No time conflicts between selected classes
- Courses will have enrollment limits
- There will be a wait-list feature that will allow the instructors to let students in when the course the instructors are assigned to is full.

Review and Rating System

Students will have the opportunity to submit reviews and ratings of the class they are enrolled in; however, students will only be able to rate the class before the instructor posts the grade.

- The rating system is assigned with though 1 to 5 stars, which 1 star means the worst and the 5 stars mean the best.
- The students who submit reviews to the class they are enrolled to will have their identities be anonymous to everyone expect the registrars.
- There will be a list of taboo words that are setup by the registrars that if found in reviews will be changed to "***".

Warning and Suspension

There will be waring and suspension for students and instructors based on:

- The students who submit reviews and have taboo words in them will receive several warnings depending on how many taboo words are in the review.
- Students who have a GPA below 2 or fail the same course twice will be terminated automatically.
- Students who have a GPA between 2 and 2.25 will receive a warning and an interview with the registrars.
- Students will receive a warning for a reckless graduation application.
- Students who have 3 warnings will be suspended for 1 semester and must pay a fine to the registrar.

- Instructors who have average rating < 2 will be warned and instructors who get 3 warnings will be suspended.
- Instructors whose class GPA is above 3.5 or below 2.5 will be questioned by the registrars and if the instructors don't give an adequate justification, they will be warned or fired.
- The student and instructor can complain about a student or instructor to the registrar, and the individual can possibly receive a warning based on the investigation.

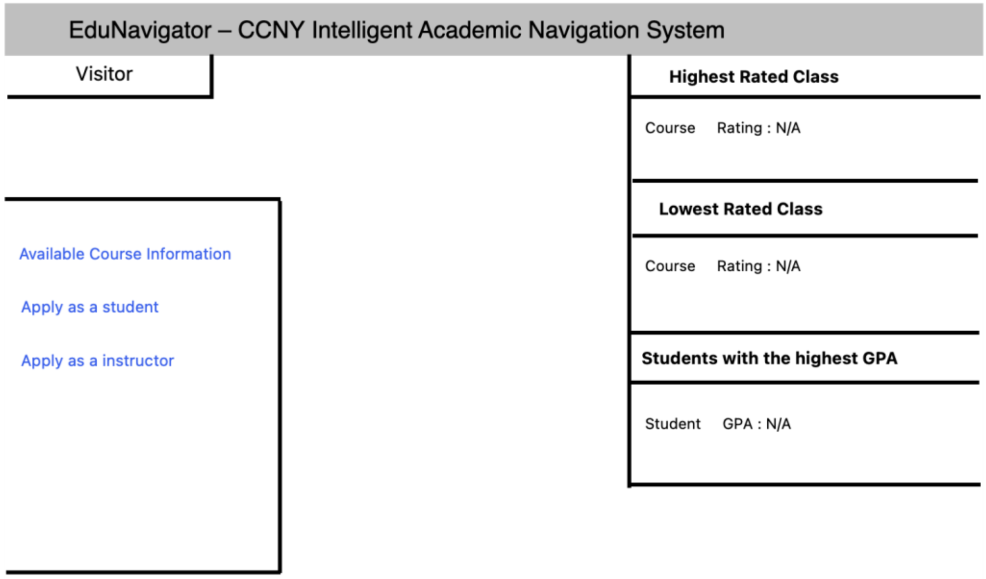
Ai Assistance

A feature that will allow the users to ask question like visitors can ask general questions about classes and requirements; students can ask additional questions on classes s/he is taking; instructors can ask additional questions about students in his/her class(es).

4. Supporting Information - David

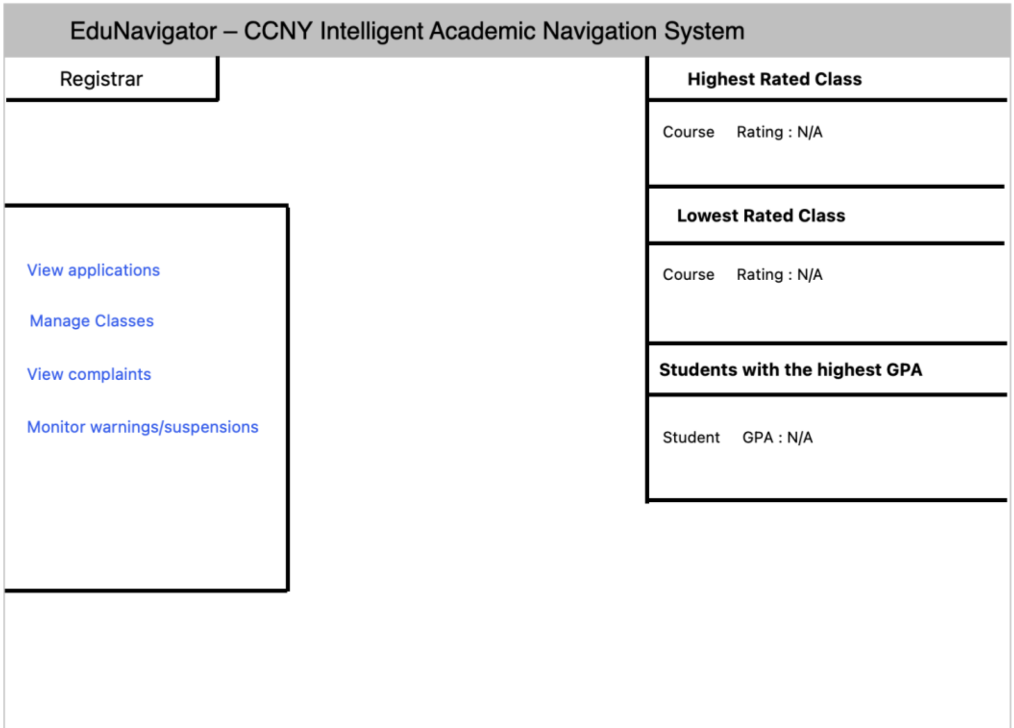
In this section, a storyboard is used to show how the system for EduNavigator will look like for the roles for visitors, registrar, instructors and students.

Visitor



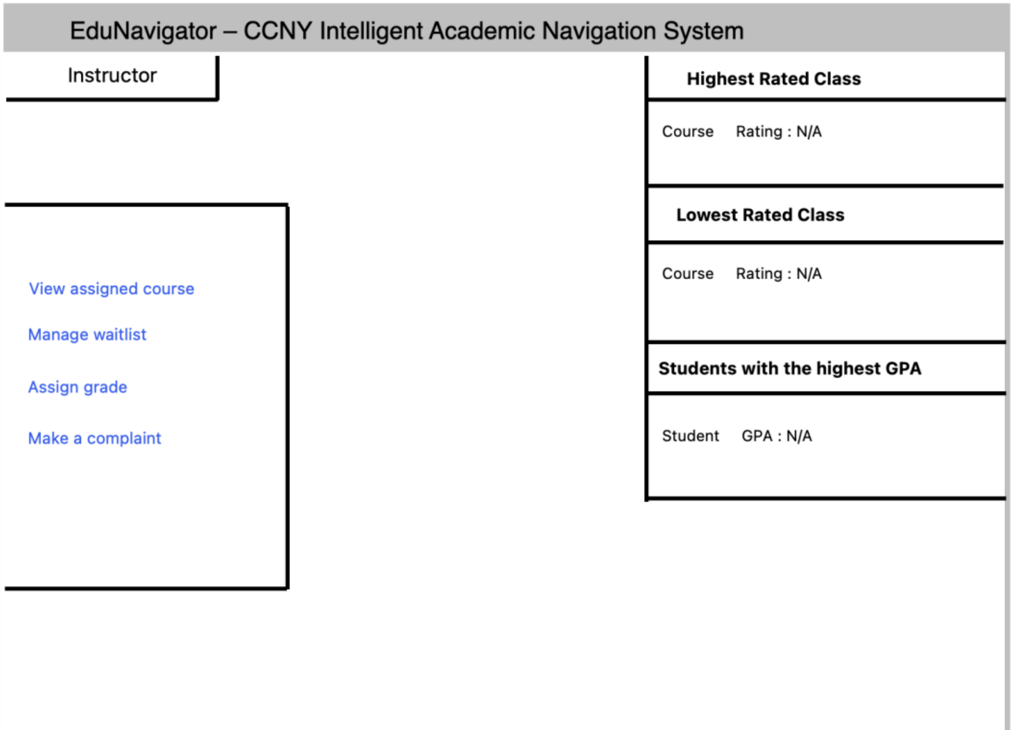
In this image, it shows the things a visitor would be able to do when logging in EduNavigator, which is able to see information about courses, and being able to apply as a student or as an instructor. The visitor would also be able to view the highest rated class, lowest rated class, and student with highest GPA on the left side of screen.

Registrar



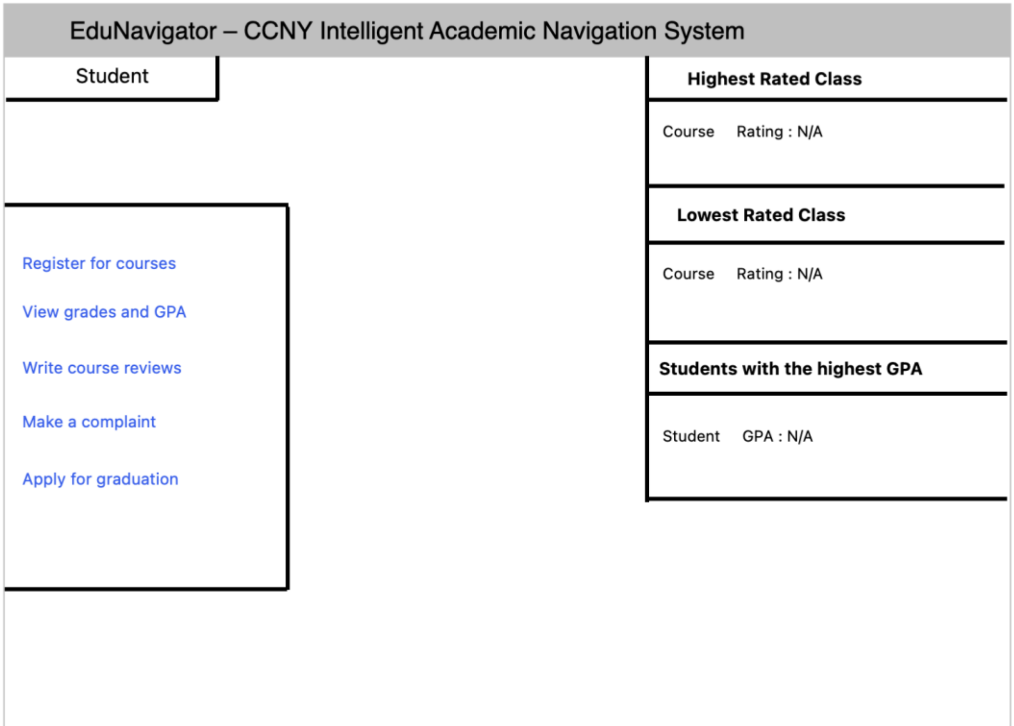
In this image, the registrar in EduNavigator would be able to see applications of visitors wanting to be students or instructors and be able to accept or reject their application if they do or don't meet the requirements.
They would be able to manage classes like assigning which instructor teaches the course and be able
to view complaints from students and instructors. Also, registrar would be able to see students and
instructors on how many warnings or suspensions they got, and they can also see the highest rated class, lowest rated class, and student with highest GPA on the left side of screen.

Instructor



The instructor would be able to do in EduNavigator is view the course they have been assigned to,
manage the waitlist of students wanting to join their course, assign grades to the student, and make a complaint to the students if they are causing an issue that will be reviewed to the registrar. The instructor
would be able to view the highest rated class, lowest rated class, and student with highest GPA on the left side of screen.

Student



In this image, the students in EduNavigator would be able to register for the courses, view their grades and GPA, write reviews on the courses they are taking, make a complaint, and be able to apply for graduation. The students would also be able to view the highest rated class, lowest rated class, and student with highest GPA on the left side of screen.